TYPES OF MOBILE CONTENT

Mobile content takes many shapes. Strictly speaking, mobile content is any form of media (pictures, music, text, videos) that can be used on a mobile device, such as a cell phone or tablet.

The demand for mobile content will continue to grow as more mobile devices arrive in the market. Devices like the iPhone, iPad and Android devices have changed how consumers consume content. Improvements in device speeds have also contributed to an increase in demand for mobile content.

Apps

Mobile app development is the act or process by which a mobile app is developed for mobile devices, such as personal digital assistants, enterprise digital assistants or mobile phones. These applications can be pre-installed on phones during manufacturing platforms, or delivered as web applications using server-side or client-side processing to provide an "application-like" experience within a Web browser.

Mobile application development, also known as mobile apps, has become a significant mobile content market since the release of the first iPhone from Apple in 2007.

The bundling of the iPhone with an app store, as well as the iPhone's unique design and user interface, helped bring a large surge in mobile application use. It also enabled additional competition from other players. For example, Google's Android platform for mobile content has further increased the amount of app content available to mobile phone subscribers.

Some examples of mobile apps would be applications to manage travel schedules, buy movie tickets, preview video content, read digital version of popular newspapers, identify music, look at star constellations, view Wikipedia, and much more.

Types of apps

Native apps

Such apps are developed for a single mobile operating system exclusively, therefore they are "native" for a particular platform or device. App built for systems like iOS, Android, Windows phone, Symbian, Blackberry cannot be used on a platform other than their own. In other words, you won't be able to use Android app on iPhone.

Hybrid apps

They are built using multi-platform web technologies (for example HTML5, CSS and JavaScript). So-called hybrid apps are mainly website applications disguised in a native wrapper. Hybrid multi-platform apps are fast and relatively easy to develop.

Web apps

Web apps use a browser to run and are usually written in HTML5, JavaScript or CSS. These apps redirect a user to URL and offer "install" option by simply creating a bookmark to their page.

Web applications require minimum of device memory, as a rule. As all personal databases are saved on a server, users can get access from any device whenever there is internet connection. That is why the use of web apps with poor connection would result in bad user experience.

Categories of apps

This clearly indicates what purposes people are using mobile apps for. They tend to do business, communicate, entertain and play games, educate themselves, and relax, of course.

Gaming apps

Mobile gaming has always been thriving, prompting app developers to invest more time and resources into creating new games and mobile versions of well-known stationary games. From

user perspective gaming apps offer the sense of achievement and high engagement in the process – an average gaming session is about 8 minutes long.

Business apps

Also referred to as productivity apps. Billing, buying, booking, sending emails, tracking working progress – you name it. Business apps vary from B2B applications to office and personal apps to boost productivity and minimize expense: imagine that you can find Android app developers for hire, recruit a new sales-manager or buy new cartridges for your office printers just with one tap.

Lifestyle apps

Probably, the widest category in 'types of apps' topic, as it covers plenty of common human activities and interests. From shopping, fitness and workout tracker to fashion, virtual fitting room and even weight loss. Such applications for personal lifestyle often are the ones that push technical innovations the most and developers constantly get requests to create a solution for personal or working tasks, or leisure and fun.

Entertainment apps

We are talking about streaming, chatting, searching events, watching videos online, posting photos to Instagram and so on.

These are the apps that have a tendency to invoke a kind of dependency, they keep us engaged, logged in, always checking for updates. On the bright side, imagination and talent of app creators bring lots of fresh and fun things to our lives, so we will never be bored again, that's for sure.

Utility apps

Utility software is used on a daily basis by literally every one of us, we don't even realize it. Though unlike others types of apps, utility apps display the shortest user session times. People use these to just get things done and move on. Falling into top categories of mobile apps can be explained by sheer amount of practical individuals leveraging their gadgets to full potential. For example, *taxi apps*. However, most popular types of applications here are scanners, trackers, healthcare, first aid manuals, etc.

Travel apps

Travel apps' purpose is to make your traveling easier, more comfortable, fun and informative. Some of them turn your smartphone into universal travel diary, some can literally guide you through the unknown sites abroad using maps, some provide translation assistance. That's why most of today's tourists seem to be digitally-savvy all-knowing travelers.

Images

Mobile images are used as the wallpaper to a mobile phone, and are also available as screensavers. On some handsets, images can also be set to display when a particular person calls the users.

Music

Mobile music is any audio file that is played on a mobile phone. Mobile music is normally formatted as an AAC (Advanced Audio Coding) file or an MP3, and comes in several different formats. Monophonic ringtones were the earliest form of ringtone, and played one tone at a time. This was improved upon with polyphonic ringtones, which played several tones at the same time so a more convincing melody could be created.

The next step was to play clips of actual songs, which were dubbed Real tones. These are preferred by record labels as this evolution of the ringtone has allowed them to gain a cut of lucrative ringtone market. In short Realtones generate royalties for record labels (the master recording owners) as well as publishers (the writers), however, when Monophonic or Polyphonic ringtones are sold only publishing or "mechanical" royalties are incurred as no master recording has been exploited.

Some companies promote covertones, which are ringtones that are recorded by cover bands to sound like a famous song. Recently Ringback tones have become available, which are played to the person calling the owner of the ringback tone. Voicetones are ringtones that play someone talking or shouting rather than music, and there are various of ringtones of natural and everyday sounds.

A recent innovation is the singtone, whereby "the user's voice is recorded singing to a popular music track and then "tuned-up" automatically to sound good. This can then be downloaded as a ringtone or sent to another user's mobile phone".

As well as mobile music there are full track downloads, which are an entire song encoded to play on a mobile phone. These can be purchased and bought over the mobile network, but data charges can make this prohibitive. The other way to get a song onto a mobile phone is by "side loading" it, which normally involves downloading the song onto a computer and then transferring it to the mobile phone via Bluetooth, infra-red or cable connections. It is possible to use a full track as a ringtone. In recent years, websites have sprung that allow users to upload audio files and customize them into ringtones using specialized applications.

Video

Mobile video comes in several forms including 3GPP, MPEG-4, RTSP, and Flash Lite.

Mobishows and cellsodes

A Mobishow or a cellsode are terms to describe a broadcast quality programme / series which has been produced, directed, edited and encoded for the mobile phone. Mobishows and Cellsodes can range from short video clips such as betting advice or the latest celebrity gossip, through to half-hour drama serials.

Streaming

Radio

Mobile streaming radio is an application that streams on-demand audio channels or live radio stations to the mobile phone. Today, all major carriers offer some sort streaming radio service featuring programmed stations based on popular genres and live stations which included both music and talk.

TV

Mobile video also comes in the form of streaming TV over the mobile network, which must be a 2.5G or 3G network. This mimics a television station in that the user cannot elect to see what they wish but must watch whatever is on the channel at the time.

There is also mobile broadcast TV, which operates like a traditional television station and broadcasts the content over a different spectrum. This frees up the mobile network to handle calls

and other data usage, and because of the "one-to-many" nature of mobile broadcast TV the video quality is a lot better than that streamed over the mobile networks, which is a "one-to-one" system.

Live video

Live video can also be streamed and shared from a cell phone through applications like Qik and InstaLively. The uploaded video can be shared to friends through emails or social networking sites. Most Live video streaming application works over the cell network or through Wi-Fi. They also require most users to have a data plan from their cell phone carriers.

Text formats

No matter what format text content is in, reading on a small screen will be much easier if the text can 'reflow', so that the reader only has to scroll down, not sideways. Reflow also makes reading easier for users who need to change the font size to suit their eyesight.

All e-book readers, internet enabled mobile phones and tablets natively support plain text and HTML. Both of these formats will reflow without any special treatment. However, formats which can be used to prevent copying of text, such as PDF, only reflow if they've been set up in a way that allows it. The reflow feature improves accessibility for screen readers as well as for mobile devices.

Phone type	PDF	HTML	Plain text	ePub
Smartphone (iPhone/Android)	Yes	Yes	Yes	Yes
Feature phone (Blackberry/ N95)	Maybe	Yes	Yes	Maybe
Basic internet phone	No	Yes	Yes	No

Text formats supported by mobile phones

ePub

ePub is a widely adopted open source standard for e-books. Most dedicated e-book readers, tablets and smartphones can render ePub well, so that text reflows to fit the width of the screen and tables and diagrams display correctly.

“EPUB is the distribution and interchange format standard for digital publications and documents based on Web Standards. EPUB defines a means of representing, packaging and encoding structured and semantically enhanced Web content — including XHTML, CSS, SVG, images, and other resources — for distribution in a single-file format. EPUB allows publishers to produce and send a single digital publication file through distribution and offers consumers interoperability between software/hardware for unencrypted reflowable digital books and other publications.”

PDF

Scholarly peer-reviewed articles are most commonly supplied in PDF format. The majority of smartphones, tablets and netbooks will support reading PDFs, although in most cases apps will need to be installed to enable this.

Many PDFs are set up in a way that doesn’t allow them to reflow, which leaves a mobile user feeling as though they are reading through a small hole, and they have to move the document around under the hole in order to read it, which can be confusing and frustrating.

Audio visual content

When serving audiovisual content to mobile devices the constraints are file size and connectivity. Allowing users to download content allows them to access it offline, but takes up space on their device and enables them to share it with others. Streaming content can help to prevent unauthorized sharing, but can increase users data costs or lead to a poor experience on a slow connection as the streaming content will buffer very slowly.

Most handheld devices support audio files in popular formats such as MP3. However, audio files can be large and due to limited storage capacity and data bandwidth users will not always be able to download files larger than a megabyte. If a file is large give users the option to access a transcript instead.

Video files can be played on handheld devices through purpose built apps and through some mobile browsers.